

PERSONAL STATEMENT

Nanyang Technological University, Singapore
MSc in Computer Control & Automation

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I am pursuing NTU's MSc in Computer Control & Automation because a real-robot deployment taught me that execution semantics determine whether a learned command remains meaningful. In an OpenPI loop for a Dobot CR5, I assembled RGB and robot-state observations, received action chunks through a WebSocket server, and translated them into ServoP arm targets and Modbus gripper commands. Under non-blocking operation, a new inference could consume an intermediate in-motion state before the previous chunk was complete, producing back-and-forth corrections. I adopted blocking, short-horizon execution, trading apparent frequency for state-action consistency. This solved the immediate problem but left me wanting the control theory to analyse it.

For the August 2027 intake, subject to course availability, EE6203 Computer Control Systems would help me examine computer-controlled dynamics instead of relying on one workaround. EE6225 Multivariable Control Systems Analysis & Design would deepen my understanding of coupled motion and feedback, while EE6221 Robotics & Intelligent Sensors would connect that analysis to the sensed state on which execution depends. These courses match the gap between building a functioning pipeline and explaining it systematically.

If I choose the coursework-only option and EE6008 Collaborative Research & Development Project is available, its team setting could support a bounded study of chunked execution. I would vary state-sampling points or completion rules, instrument the motion, and compare observations against a control model. I would contribute experience with synchronized records, policy serving, and real-hardware integration while learning to define, prototype, test, and communicate with other engineers. My goal is robotics or embodied-AI R&D connecting learning with control and dependable deployment. NTU would help me treat timing, feedback, and physical motion as one engineering problem.